A nine-vertex counterexample to Jensen–Toft Problem 5.8

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October 19, 2026

Abstract

We give a negative answer to the question in Jensen and Toft, *Graph Coloring Problems*, Section 5.8, asking whether every noncomplete k-critical graph, for $k \geq 4$, contains a noninduced $(k-1)$-critical subgraph. The witness is the classical nine-vertex 5-Ore graph O_5. We show that its 4-critical subgraphs are exactly three copies of K_4 and twelve copies of the Moser spindle, all induced. The proof is elementary. A Lean formalization independently certifies the counterexample property.

1 The question

All graphs are finite and simple. A graph is *k-critical* if its chromatic number is k and every proper subgraph is $(k-1)$-colorable. A subgraph H of G is *induced* if $H = G[V(H)]$.

Jensen and Toft [1, Section 5.8, p. 103] ask whether every noncomplete k-critical graph, for $k \geq 4$, contains a $(k-1)$-critical subgraph that is not induced. We answer this question as stated.

Theorem 1. *There is a noncomplete 5-critical graph G on nine vertices and nineteen edges whose 4-critical subgraphs are exactly fifteen induced subgraphs: three copies of K_4 and twelve copies of the Moser spindle. In particular, G has no noninduced 4-critical subgraph.*

The host is the classical graph commonly denoted O_5. We do not claim a new construction of the host graph. Our argument describes its 4-critical subgraphs and applies that description to the question above.

2 The graph and its criticality

Use the vertex set $\{0, \dots, 8\}$, and write

$$L = \{0, 1\}, \qquad R = \{4, 5\}, \qquad Z = \{2, 3, 7\}, \qquad u = 6, \qquad v = 8.$$

The set $L \cup R$ induces K_4, and Z induces K_3. Both u and v are adjacent to every vertex of Z. In addition, u is adjacent to every vertex of R, and v to every vertex of L. There are no other edges. In particular, uv is not an edge.

This is an Ore composition of two copies of K_5: delete the edge uv from one copy, split a vertex of the other copy into two vertices with disjoint two-element neighborhoods, and identify the split vertices with u and v. The explicit description above gives $6 + 3 + 6 + 4 = 19$ edges.

Lemma 1. *The graph G is noncomplete and 5-critical.*

Proof. Suppose that G has a proper four-coloring. The clique $L \cup R$ uses all four colors. The color of u must occur on L, and the color of v must occur on R. Thus u and v have different colors. Every vertex of the triangle Z is adjacent to both, leaving only two colors for Z, a contradiction.

A proper five-coloring, in vertex order $0, \ldots, 8$, is

$$(0, 1, 0, 1, 2, 3, 4, 2, 3).$$

It remains to check criticality. Permutations within L, within R, and within Z are automorphisms. Exchanging L with R and u with v is also an automorphism. These automorphisms give five edge orbits, represented by the edges in Table 1. Each row supplies a proper four-coloring after deletion of that edge.

Deleted edge	Four-coloring in vertex order $0, \ldots, 8$
01	$(0, 0, 3, 2, 3, 2, 1, 0, 1)$
04	$(0, 3, 3, 2, 0, 2, 1, 0, 1)$
23	$(0, 2, 0, 0, 3, 1, 2, 3, 1)$
26	$(0, 2, 0, 3, 3, 1, 0, 2, 1)$
46	$(0, 3, 3, 2, 1, 2, 1, 0, 1)$

Table 1: Four-colorings of representative edge deletions.

Every edge deletion is therefore four-colorable. Every vertex has an incident edge, so every vertex deletion is four-colorable as well, by restricting a coloring of an incident-edge deletion. Every proper subgraph is contained in an edge or vertex deletion. Finally, u and v are nonadjacent, so G is not complete. □

3 The 4-critical subgraphs

We first record two elementary facts.

Lemma 2. *Every graph on at most five vertices that is not three-colorable contains K_4.*

Proof. Suppose otherwise, and take a counterexample with the fewest vertices. Every vertex has degree at least three: otherwise a three-coloring after its deletion extends to the whole graph. A counterexample on four vertices would be K_4, so there are five vertices. Its complement has maximum degree at most one; its missing edges therefore form a matching. If there is at most one missing edge, the graph contains K_4. If there are two, give each nonadjacent pair one color and the remaining vertex a third. Both possibilities give a contradiction. □

Lemma 3. *For every three-element subset T of $L \cup R$ and every two-element subset P of Z, the induced graph*

$$G[T \cup P \cup \{u, v\}]$$

is a 4-critical Moser spindle.

Proof. By symmetry, write $T = \{a, b, c\}$ with $a, b \in L$ and $c \in R$, and write $P = \{x, y\}$. The graph consists of two diamonds (copies of K_4 with one edge deleted), sharing the vertex v, together with one additional edge. The first diamond has vertices $\{a, b, c, v\}$ and nonadjacent terminals c, v. The

second has vertices $\{x, y, u, v\}$ and nonadjacent terminals u, v. The additional edge is cu. This is the usual seven-vertex, eleven-edge description of the Moser spindle.

In every three-coloring of a diamond, its terminals have the same color. The two diamonds would therefore force c, v, u to have the same color, contradicting the edge cu.

For completeness, every edge deletion is three-colorable. Deleting cu permits both diamonds to be colored with all three terminals equal. If an edge of a diamond is deleted, that diamond admits a three-coloring with different terminal colors. To check this last assertion, name the terminals p, q and the other vertices r, s. After deleting rs, use $(p, q, r, s) = (0, 1, 2, 2)$. After deleting pr, use $(p, q, r, s) = (0, 1, 0, 2)$. Symmetry covers the other three edges.

Thus, after deleting an edge of either diamond, color its terminals differently and color the other diamond with equal terminals. Color permutations make the colorings agree at v; the edge cu then has different endpoint colors. A three-coloring of an edge deletion also gives a four-coloring of the original graph by assigning one endpoint a fourth color. Every vertex has positive degree, so the edge-deletion checks imply colorability of every proper subgraph. □

Proof of Theorem 1. Lemma 1 establishes the required properties of the host. Its copies of K_4 are exactly those induced by

$$L \cup R, \qquad Z \cup \{u\}, \qquad Z \cup \{v\}.$$

There are no edges between $L \cup R$ and Z, and u, v are nonadjacent, so there are no other copies. Lemma 3 gives twelve further induced 4-critical subgraphs, since there are $\binom{4}{3}\binom{3}{2} = 12$ choices of (T, P). We show that this list is exhaustive.

Let F be a 4-critical subgraph of G. If F contains K_4, criticality forces F to equal that K_4. Assume henceforth that F contains no K_4.

For the coloring argument, add any missing host vertices as isolated vertices. Split the edges of F into a graph X on $(L \cup R) \cup \{u, v\}$ and a graph Y on $Z \cup \{u, v\}$. Their common vertex set consists of the independent terminals u, v. By Lemma 2, Y is three-colorable.

The graph X has a three-coloring with u and v different. Indeed, some edge ab is missing from its four-vertex clique side. Color $K_4 - ab$ on $L \cup R$ by giving a, b color 0 and the other two vertices colors $1, 2$. If a, b lie in different parts, give u the color of the remaining vertex of L and v the color of the remaining vertex of R. If $a, b \in L$, give u color 0 and v color 1; if $a, b \in R$, reverse these terminal assignments. These colorings respect all permitted terminal attachments and hence all edges of X.

Two three-colorings of X and Y can be joined after a color permutation whenever they give the terminal pair the same equality relation. Since F is not three-colorable and X permits different terminal colors, every three-coloring of Y must give u, v the same color. Consequently $Y + uv$ is not three-colorable. Lemma 2 gives a K_4 in $Y + uv$. It must use the added edge, because Y contains no K_4. Thus Y contains the diamond on u, v and some pair $P \subset Z$.

Since Y is three-colorable with equal terminal colors, X cannot admit such a coloring. Identifying u and v in X therefore gives a non-three-colorable graph on five vertices. By Lemma 2, it contains K_4. This K_4 must contain the identified terminal, since the clique side of X contains no K_4. It follows that X contains a triangle on some three-element subset $T \subset L \cup R$, together with the permitted terminal attachment of each of its vertices. The uniqueness of each vertex's permitted terminal neighbor justifies lifting these three edges from the identified graph.

The triangle and these attachments, together with the diamond in Y, form precisely $G[T \cup P \cup \{u, v\}]$. Lemma 3 shows that this is 4-critical. Criticality of F forces equality. The fifteen listed subgraphs are distinct and induced, proving the theorem. □

4 Formal verification

The Lean source and reproducible build instructions are at

`https://github.com/alexgocardinal/jt58-critical-subgraphs`.

The repository is private during manuscript preparation.

The formal theorem `JT58.jt58_counterexample` proves 5-criticality with respect to arbitrary proper subgraphs, noncompleteness, and the absence of a noninduced 4-critical subgraph. The formalization uses Lean 4.35.0 and does not require Mathlib. Explicit certificates check the edge and vertex deletions of the host. For the remaining assertion, a direct case proof covers all 2^{19} edge subsets using fifteen induced non-three-colorable cores and 264 three-coloring certificates. Its 2,986 terminal branches are checked by Lean's kernel.

Python scripts generate proof files, but their computations are not trusted premises. The checked-in Lean files suffice for verification. The axiom audit of the final theorem reports exactly

`propext`, `Classical.choice`, `Quot.sound`.

The proof uses neither a placeholder axiom nor a native-computation axiom. Running `lake build` checks the formal counterexample theorem. The exact classification in Theorem 1 is established by the elementary argument above.

References

[1] T. R. Jensen and B. Toft, *Graph Coloring Problems*, Wiley-Interscience, New York, 1995. Chapter 5.